

GeekGyan

The majority of computer users blink 7 times per minute as compared to a normal blinking rate of 20 times a minute.

Tech woes

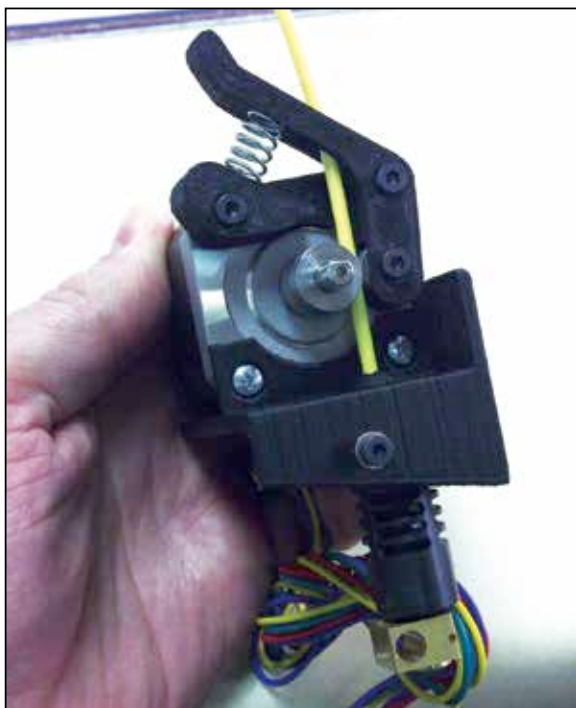
Tech Giants ask POTUS to resist backdoor access to encrypted data.
<http://dgit.in/1B9llYh>

slightly more complex. If you are dedicating all your time to it, then it's a weekend project. After building it, you will need some time to get used to it.

Kit for Prusa i3: <http://www.3dprintronics.com/> provides good instructional videos to learn the assembly process, or you can procure your own parts. If you are doing the latter, I will suggest using the EiNSTeiN variant, a guide for it can be found on <http://dgit.in/1Fy08uU> and general documentation on <http://dgit.in/1Aod5Zx>.

3D Printing Components can be sourced from the following sources:

- <http://makemendel.com/>



One of the more important control heads that you'll need



Your 3D printer in action

- <http://www.3dprintronics.com/>
- <http://www.diy-india.com/>

The single plate version is the most popular one and comes highly recommended. You can get an aluminum or acrylic plate. Just a heads up: the latter is a bit weaker and there is a chance of cracking if unnecessary force is applied.

Electronics

Ramps 1.4 and Printerboard are the electronics predominantly available in India. Printerboard is more expensive but much simpler and compact to use and generally doesn't fail.

Ramps 1.4 is modular but has some faults in its design which might cause breakdown. I recommend the former, as it will give you less trouble in the long run. It's simply plug-and-play and it takes about half an hour to do all the wiring the first time around.

I started out with Ramps 1.4 but now prefer the simplicity of Printerboard.

LCD

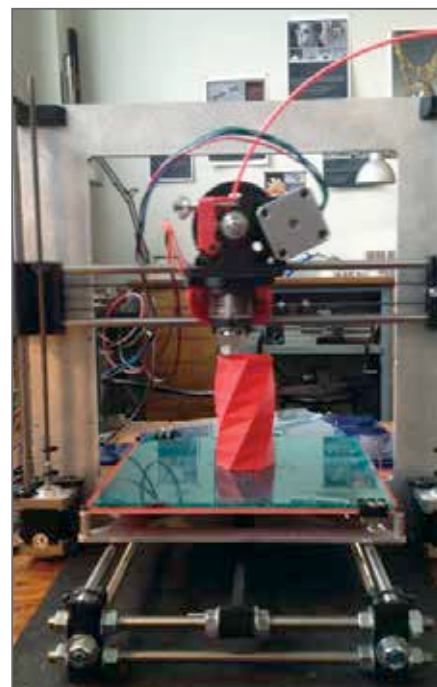
The LCD allows you to control your printer without having to use a computer. It's essential to have one of these because some prints can take as long as 20 hours to emerge, forcing you to keep your computer switched on for the entire duration.

And if the computer hangs or undergoes updates, it will cause the printing to fail. So buy an LCD when building a 3D printer.

Both Ramps and Printerboard have different LCD modules available, which are generally available at stores which sell the electronics themselves. It's a simple add-on, just enable the firmware and you're set.

Extruder

The extruder is the backbone of the 3D printer, for it does



This is how your completed 3D printer might look

the job of melting the plastic and extruding it through a small nozzle. A 0.4mm sized nozzle is a common and safe choice. There are two parts to the extruder – use the 1.75mm versions of both. (1.75mm, is the diameter of the filament).

There have been rare cases in which the older, 3.0mm versions had to be used.

Hot-End: Building the hot-end unit will take about 2 hours. Hot-End refers literally to the parts which heat up. A J-Head design here is a good choice and widely available. Don't go for metallic versions as finding a good one is difficult and they are also expensive. Consider this particular part as a consumable one, which means its performance quality will go down after a while, depending on the usage. In my case, most hot-ends have worked for about 6 months or even a year. Pro-tip? Prolong its life by keeping the filament free from dust and using the attached sponge to keep it clean.

Cold-End: Direct Drive extruders are highly recommended. Today, aluminum versions are also available which are heavier than the former but show better performance and don't break. I suggest you go for this design: <http://dgit.in/>